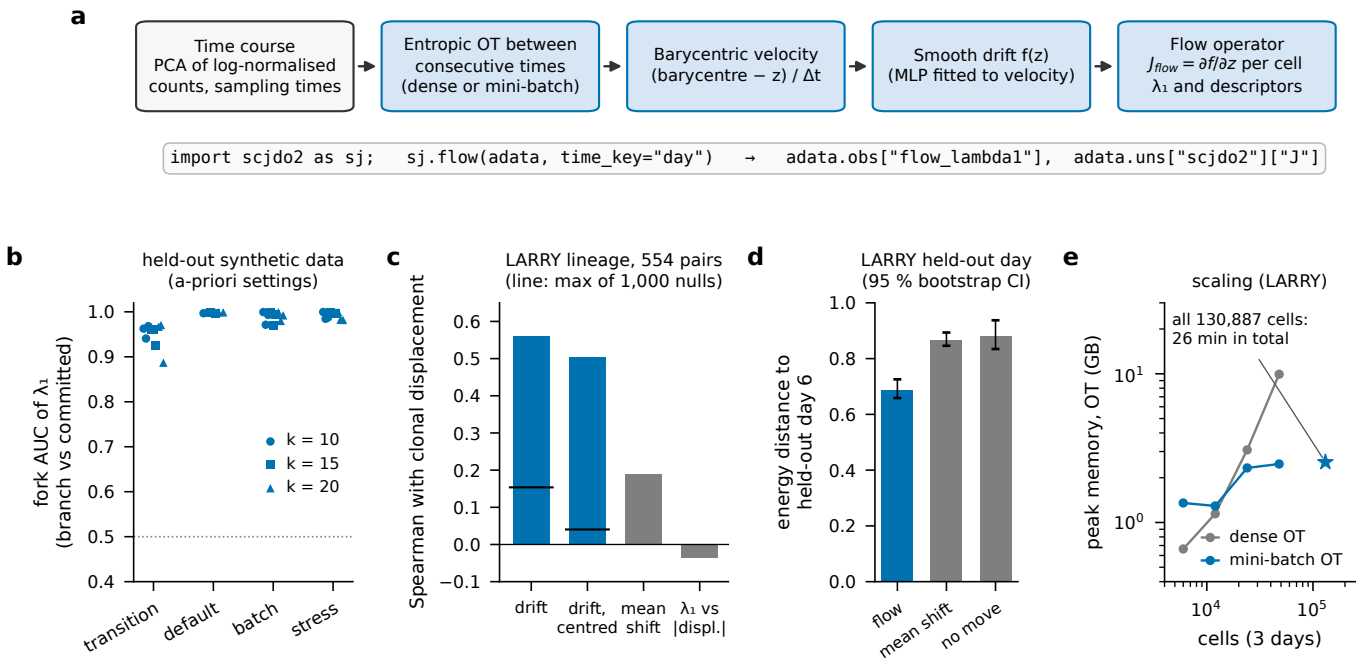
a
Time course
PCA of log-normalised
counts, sampling times
Entropic OT between
consecutive times
(dense or mini-batch)
Barycentric velocity
(barycentre − z) / Δt
Smooth drift f(z)
(MLP fitted to velocity)
Flow operator
$J_{flow} = \partial f/\partial z$ per cell
λ_1 and descriptors
import scjdo2 as sj; sj.flow(adata, time_key="day") → adata.obs["flow_lambda1"], adata.uns["scjdo2"]["J"]
b
held-out synthetic data
(a-priori settings)
fork AUC of λ_1
(branch vs committed)
k = 10
k = 15
k = 20
transition
default
batch
stress
c
LARRY lineage, 554 pairs
(line: max of 1,000 nulls)
Spearman with clonal displacement
drift
drift, centred
mean shift
λ_1 vs |displ.|
d
LARRY held-out day
(95 % bootstrap CI)
energy distance to held-out day 6
flow
mean shift
no move
e
scaling (LARRY)
all 130,887 cells:
26 min in total
peak memory, OT (GB)
dense OT
mini-batch OT
cells (3 days)